



SERVEI DHCP AMB KEA

NF1.1_PR201



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Assignació del rang IP:

Nº de llista: 63

IP server (kea-server): 172.24.63.10

Pool dinàmic: 172.24.63.100 - 172.24.63.200

Reserves per MAC: 172.24.63.50, 172.24.63.51, 172.24.63.52

Configuració netplan: xarxa enp0s3 amb DHCP desactivat i la IP 172.24.63.10/24

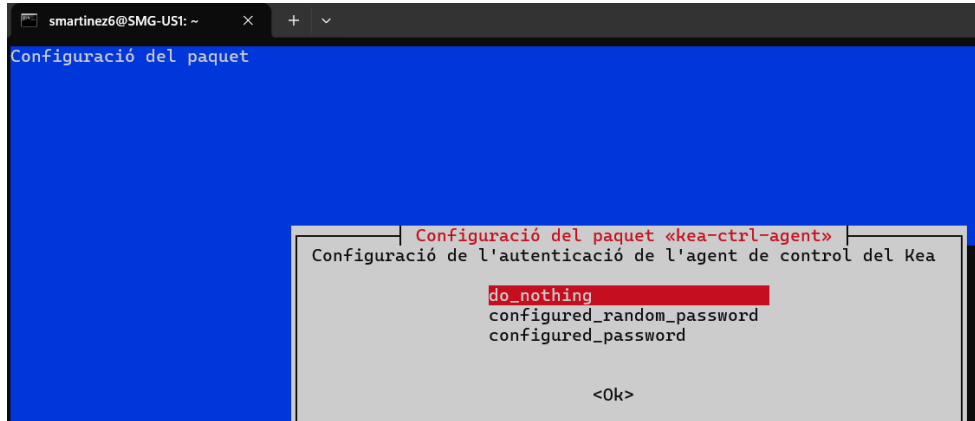
```
GNU nano 8.7.1 /etc/netplan/00-installer-config.yaml
# This is the network config written by 'subiquity'
network:
  ethernets:
    enp0s3:
      dhcp4: false
      dhcp6: true
      match:
        macaddress: 08:00:27:d4:73:fa
      set-name: enp0s3
      addresses: [172.24.63.10/24]
    enp0s8:
      dhcp4: true
  version: 2
```

Verificació de la IP estàtica indicada i en estat Up.

```
smartinez6@kea-server-shailamartinez:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP group default qlen 1000
    link/ether 08:00:27:d4:73:fa brd ff:ff:ff:ff:ff:ff
    altname enx080027d473fa
    inet 172.24.63.10/24 brd 172.24.63.255 scope global enp0s3
        valid_lft forever preferred_lft forever
    inet6 fe80::a00:27ff:fed4:73fa/64 scope link proto kernel_ll
        valid_lft forever preferred_lft forever
```

Part 1: Instal·lació de Kea

Durant la instal·lació de Kea surt una pantalla de gestor de paquets dpkg, selecciono l'opció per defecto "do_nothing" per saltar la creació de contrasenya per fer les proves de la pràctica.



Hostname antic: SMG-US1, Hostname actual: kea-server-shailamartinez

Comprovació del estat del kea-dhcp4-server i kea-ctrl-agent en Enabled i en execució (Active running).

```
smartinez6@kea-server-shailamartinez:~$ sudo hostnamectl set-hostname kea-server-shailamartinez
smartinez6@kea-server-shailamartinez:~$ hostname
kea-server-shailamartinez
smartinez6@kea-server-shailamartinez:~$ systemctl status kea-dhcp4-server
● kea-dhcp4-server.service - Kea IPv4 DHCP daemon
   Loaded: loaded (/usr/lib/systemd/system/kea-dhcp4-server.service; enabled; preset: enabled)
   Active: active (running) since Sun 2026-09-20 17:37:12 UTC; 5min ago
     Invocation: 80bf3df764124f7ba8287787b0df5119
       Docs: man:kea-dhcp4(8)
    Main PID: 4136 (kea-dhcp4)
      Tasks: 7 (limit: 1891)
     Memory: 2.4M (peak: 2.8M)
        CPU: 52ms
     CGroup: /system.slice/kea-dhcp4-server.service
             └─4136 /usr/sbin/kea-dhcp4 -c /etc/kea/kea-dhcp4.conf
smartinez6@kea-server-shailamartinez:~$ systemctl status kea-ctrl-agent
● kea-ctrl-agent.service - Kea Control Agent
   Loaded: loaded (/usr/lib/systemd/system/kea-ctrl-agent.service; enabled; preset: enabled)
   Active: active (running) since Sun 2026-09-20 17:38:53 UTC; 3min 35s ago
     Invocation: 7995d5580ed8475191a7753c5748b2d5
       Docs: man:kea-ctrl-agent(8)
    Main PID: 4241 (kea-ctrl-agent)
      Tasks: 5 (limit: 1891)
     Memory: 1.7M (peak: 2.2M)
        CPU: 25ms
     CGroup: /system.slice/kea-ctrl-agent.service
             └─4241 /usr/sbin/kea-ctrl-agent -c /etc/kea/kea-ctrl-agent.conf
smartinez6@kea-server-shailamartinez:~$
```

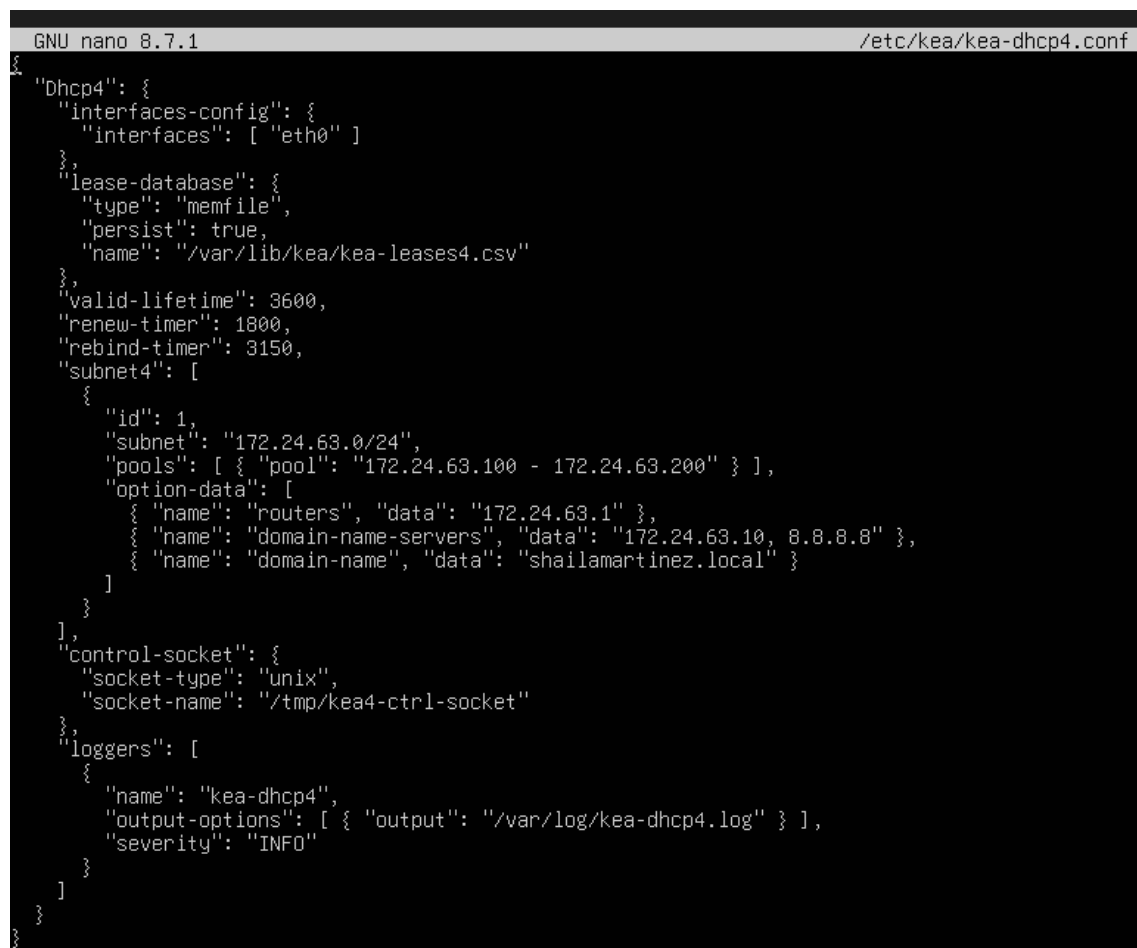
Pregunta documentació: per què Kea és el substitut recomanat d'ISC DHCP i quin avantatge té la seva configuració en format JSON respecte al format text tradicional? → Kea ofereix una arquitectura més moderna amb API REST, alta disponibilitat i suport per a bases de dades.

La configuració en JSON és més estructurada i fàcil de processar automatitzar que el format de text tradicional de ISC DHCP.

Part 2: Configuració del Subnet i Concessions Dinàmiques

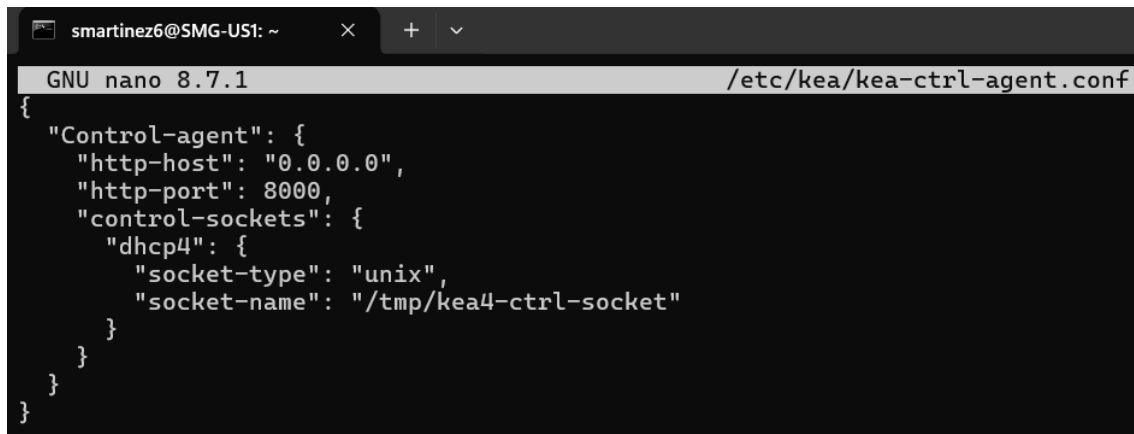
Una vegada instal·lat accedir al fitxer de configuració `etc/kea/kea-dhcp4.conf`. Indiquem la següent configuració, en el meu cas he posat l'interfície `enp0s3`. La BD de concessions "memfile" guardat a `/var/lib/kea/kea-leases4.csv`. El subnet `172.24.63.0/24`, pool dinàmic del rang `172.24.63.100` a `172.24.63.200`.

A les opcions globals: router `172.24.63.1`, el DNS `172.24.63.10 8.8.8.8` i el nom de domini `shailamartinez.local`. Control socket: `/run/kea/kea4-ctrl-socket`



```
GNU nano 8.7.1 /etc/kea/kea-dhcp4.conf
{
  "dhcp4": {
    "interfaces-config": {
      "interfaces": [ "eth0" ]
    },
    "lease-database": {
      "type": "memfile",
      "persist": true,
      "name": "/var/lib/kea/kea-leases4.csv"
    },
    "valid-lifetime": 3600,
    "renew-timer": 1800,
    "rebind-timer": 3150,
    "subnet4": [
      {
        "id": 1,
        "subnet": "172.24.63.0/24",
        "pools": [ { "pool": "172.24.63.100 - 172.24.63.200" } ],
        "option-data": [
          { "name": "routers", "data": "172.24.63.1" },
          { "name": "domain-name-servers", "data": "172.24.63.10, 8.8.8.8" },
          { "name": "domain-name", "data": "shailamartinez.local" }
        ]
      }
    ]
  },
  "control-socket": {
    "socket-type": "unix",
    "socket-name": "/tmp/kea4-ctrl-socket"
  },
  "loggers": [
    {
      "name": "kea-dhcp4",
      "output-options": [ { "output": "/var/log/kea-dhcp4.log" } ],
      "severity": "INFO"
    }
  ]
}
```

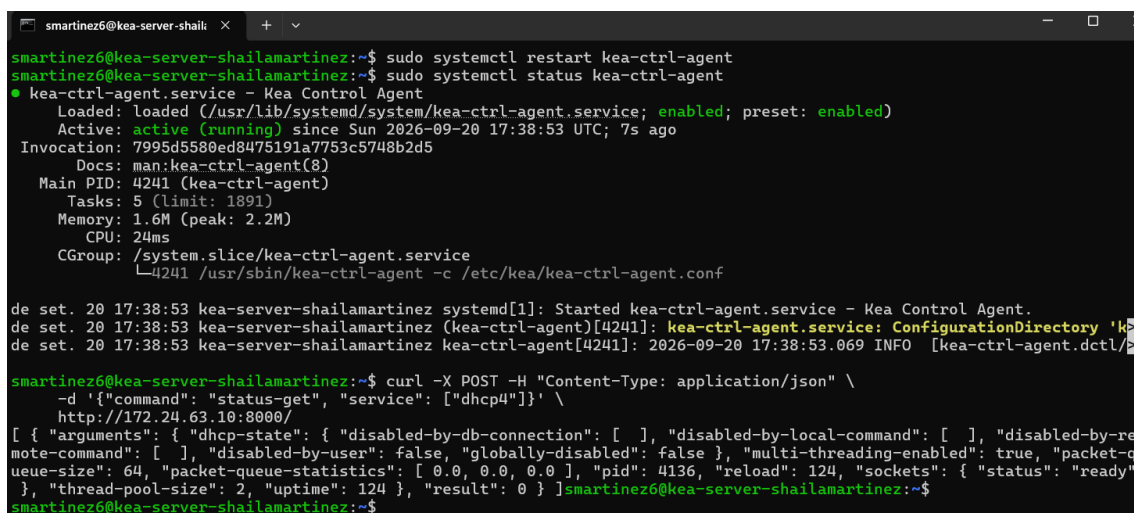
Fitxer de configuració control agent (`/etc/kea/kea-ctrl-agent.conf`), escolta el `0.0.0.0` i el port `8000` que es comunica amb el servei DHCP4 a través del socket Unix en `/run/kea/kea4-ctrl-socket`



```
GNU nano 8.7.1 /etc/kea/kea-ctrl-agent.conf
{
  "Control-agent": {
    "http-host": "0.0.0.0",
    "http-port": 8000,
    "control-sockets": {
      "dhcp4": {
        "socket-type": "unix",
        "socket-name": "/tmp/kea4-ctrl-socket"
      }
    }
  }
}
```

Reinici del servei kea-ctrl-agent i després es verifica que estigui actiu i en execució. Després consulta via API REST (status-get) amb la comanda de curl...

La resposta JSON torna "result": 0 i l'estructura "dhcp-state", confirmant que el servei DHCPv4 està preparat (ready) i responent mitjançant l'API.



```
smartinez6@kea-server-shaila:~$ sudo systemctl restart kea-ctrl-agent
smartinez6@kea-server-shaila:~$ sudo systemctl status kea-ctrl-agent
● kea-ctrl-agent.service - Kea Control Agent
   Loaded: loaded (/usr/lib/systemd/system/kea-ctrl-agent.service; enabled; preset: enabled)
   Active: active (running) since Sun 2026-09-20 17:38:53 UTC; 7s ago
     Invocation: 7995d5580ed8475191a7753c5748b2d5
       Docs: man:kea-ctrl-agent(8)
    Main PID: 4241 (kea-ctrl-agent)
      Tasks: 5 (limit: 1891)
     Memory: 1.6M (peak: 2.2M)
        CPU: 24ms
    CGroup: /system.slice/kea-ctrl-agent.service
            └─4241 /usr/sbin/kea-ctrl-agent -c /etc/kea/kea-ctrl-agent.conf

de set. 20 17:38:53 kea-server-shailamartinez systemd[1]: Started kea-ctrl-agent.service - Kea Control Agent.
de set. 20 17:38:53 kea-server-shailamartinez (kea-ctrl-agent)[4241]: kea-ctrl-agent.service: ConfigurationDirectory 'k
de set. 20 17:38:53 kea-server-shailamartinez kea-ctrl-agent[4241]: 2026-09-20 17:38:53.069 INFO [kea-ctrl-agent.dctl/

smartinez6@kea-server-shaila:~$ curl -X POST -H "Content-Type: application/json" \
-d '{"command": "status-get", "service": ["dhcp4"]}' \
http://172.24.63.10:8000/
[ { "arguments": { "dhcp-state": { "disabled-by-db-connection": [ ], "disabled-by-local-command": [ ], "disabled-by-re
mote-command": [ ], "disabled-by-user": false, "globally-disabled": false }, "multi-threading-enabled": true, "packet-q
ueue-size": 64, "packet-queue-statistics": [ 0.0, 0.0, 0.0 ], "pid": 4136, "reload": 124, "sockets": { "status": "ready"
}, "thread-pool-size": 2, "uptime": 124 }, "result": 0 } ]smartinez6@kea-server-shaila:~$
```

Resposta JSON de status-get mostrant el servei actiu:

Hem comprovat l'estat del servei DHCP mitjançant API REST del control agent amb la comanda status-get. La resposta JSON confirma que el servei DHCP4 està actiu i funcionant correctament.

“Result 0”: indica que s’ha executat bé, “text” que s’ha comunicat amb dhcp4 pel socket, “arguments.pid” mostra el ID del procés kea-dhcp4 al SO, “arguments.uptime” temps en segons que el servei porta funcionant.

Part 3: Reserves d'Adreça per MAC

Es crea un bloc de reserves en /etc/kea/kea-dhcp4.conf. Edito “reservations” dins de la subxarxa 172.24.63.0/24.

MAC 08:00:27:11:22:33 IP 172.24.63.50 (servidor-impressio-shailamartinez).

MAC 08:00:27:44:55:66 IP 172.24.63.51 (pc-direccio-shailamartinez).

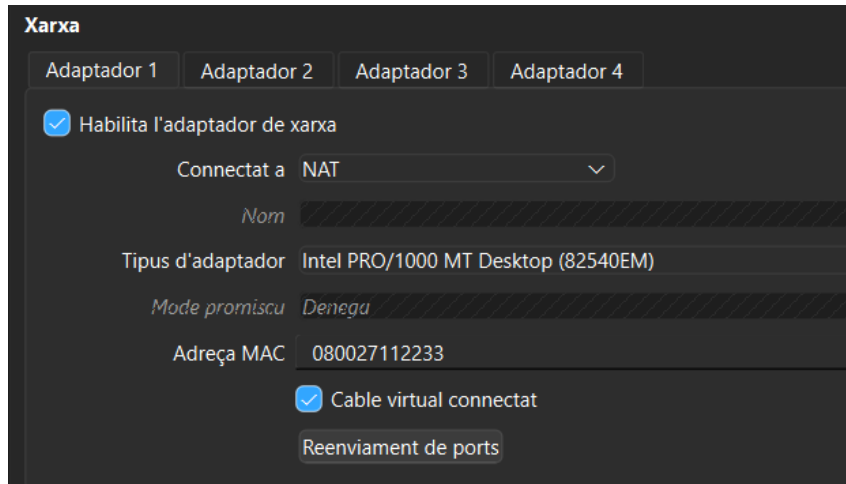
MAC 08:00:27:77:88:99 IP 172.24.63.52 (servidor-backup-shailamartinez).

```
smartinez6@kea-server-shaila: X + v
GNU nano 8.7.1 /etc/kea/kea-dhcp4.conf
},
"valid-lifetime": 3600,
"renew-timer": 1800,
"rebind-timer": 3150,
"subnet4": [
{
  "id": 1,
  "subnet": "172.24.63.0/24",
  "pools": [ { "pool": "172.24.63.100 - 172.24.63.200" } ],
  "reservations": [
    {
      "hw-address": "08:00:27:11:22:33",
      "ip-address": "172.24.63.50",
      "hostname": "servidor-impressio-shailamartinez"
    },
    {
      "hw-address": "08:00:27:44:55:66",
      "ip-address": "172.24.63.51",
      "hostname": "pc-direccio-shailamartinez"
    },
    {
      "hw-address": "08:00:27:77:88:99",
      "ip-address": "172.24.63.52",
      "hostname": "servidor-backup-shailamartinez"
    }
  ],
  "option-data": [
    { "name": "routers", "data": "172.24.63.1" },
    { "name": "domain-name-servers", "data": "172.24.63.10, 8.8.8.8" },
    { "name": "domain-name", "data": "shailamartinez.local" }
  ]
},
],
"control-socket": {
```

Verificació de configuració activa via API REST (config-get). Mostra la resposta completa del server Kea en format JSON, confirma que la configuració conté les subxarxes, opcions i bloc de reserves per MAC.

```
smartinez6@kea-server-shailamartinez:~$ sudo systemctl restart kea-dhcp4-server
smartinez6@kea-server-shailamartinez:~$ curl -X POST -H "Content-Type: application/json" -d '{"command": "config-get", "service": ["dhcp4"]}' http://172.24.63.10:8000/
{"arguments": {"Dhcp4": {"allocator": "iterative", "authoritative": false, "boot-file-name": "", "cache-threshold": 0.25, "calculate-tee-times": false, "control-sockets": [{"socket-name": "/run/kea/kea-dhcp4-ctrl-socket", "socket-type": "unix"}], "ddns-conflict-resolution-mode": "check-with-dhcid", "ddns-generated-prefix": "myhost", "ddns-override-client-update": false, "ddns-override-no-update": false, "ddns-qualifying-suffix": "", "ddns-replace-client-name": "never", "ddns-send-updates": true, "ddns-update-on-renew": false, "decline-probation-period": 86400, "dhcp-ddns": {"enable-updates": false, "max-queue-size": 1024, "ncr-format": "JSON", "ncr-protocol": "UDP", "sender-ip": "0.0.0.0", "sender-port": 0, "server-ip": "127.0.0.1", "server-port": 53001, "dhcp-client-id": true, "expired-leases-processing": {"flush-reclaimed-timer-wait-time": 25, "hold-reclaimed-time": 3600, "max-reclaim-leases": 100, "max-reclaim-time": 250, "reclaim-timer-wait-time": 10, "unwarned-reclaim-cycles": 5}, "hooks-libraries": [], "host-reservation-identifiers": [{"hw-address", "duid", "client-id"}, {"name": "char-set": "A-Za-z0-9-"}, {"interfaces-config": {"interfaces": [{"enp0s3"}], "re-detect": true}, {"ip-reservations-unique": true, "lease-database": {"name": "/var/lib/kea/kea-leases.csv", "persist": true, "type": "memfile"}, "loggers": [{"debug-level": 0, "name": "kea-dhcp4", "output-options": [{"flush": true, "maxsize": 10240000, "maxver": 1, "output": "/var/log/kea/kea-dhcp4.log", "pattern": ""}], "severity": "INFO"}], "match-client-id": true, "multi-threading": {"enable-multi-threading": true, "packet-queue-size": 64, "thread-pool-size": 0}, "next-server": "0.0.0.0", "option-data": [], "option-def": [], "parked-packet-limit": 256, "rebind-timer": 3150, "renew-timer": 1800, "reservations-global": false, "reservations-in-subnet": true, "reservations-lookup-first": false, "reservations-out-of-pool": false, "sanity-checks": {"extended-info-checks": {"fix": "lease-checks", "warn": true}, "server-hostname": "", "server-tag": "", "shared-networks": [], "stash-agent-options": false, "statistic-default-sample-age": 0, "statistic-default-sample-count": 20, "store-extended-info": false, "subnet": [{"dhcp4-interface": "", "dhcp4-interface-id": "", "dhcp4-subnet": "", "allocator": "iterative", "cache-threshold": 0.25, "calculate-tee-times": false, "id": 1, "max-valid-lifetime": 3600, "main-valid-lifetime": 3600, "option-data": [{"always-send": false, "code": 3, "csv-format": true, "data": "172.24.63.1", "name": "routers", "never-send": false, "space": "dhcp4"}, {"always-send": false, "code": 6, "csv-format": true, "data": "172.24.63.10, 8.8.8.8", "name": "domain-name-servers", "never-send": false, "space": "dhcp4"}, {"always-send": false, "code": 15, "csv-format": true, "data": "shailamartinez.local", "name": "domain-name", "never-send": false, "space": "dhcp4"}], "pools": [{"option-data": [], "pool": "172.24.63.100-172.24.63.200"}], "rebind-timer": 3150, "relay": {"ip-addresses": []}, "renew-timer": 1800, "reservations": [{"boot-file-name": "", "client-classes": [], "hostname": "servidor-impressio-shailamartinez", "hw-address": "08:00:27:11:22:33", "ip-address": "172.24.63.50", "next-server": "0.0.0.0", "option-data": [], "server-hostname": "", "client-classes": [], "boot-file-name": "", "hostname": "pc-direccio-shailamartinez", "hw-address": "08:00:27:44:55:66", "ip-address": "172.24.63.51", "next-server": "0.0.0.0", "option-data": [], "server-hostname": "", "boot-file-name": "", "client-classes": [], "hostname": "servidor-backup-shailamartinez", "hw-address": "08:00:27:77:88:99", "ip-address": "172.24.63.52", "next-server": "0.0.0.0", "option-data": [], "server-hostname": "", "store-extended-info": false, "subnet": "172.24.63.0/24", "tl-percent": 0.5, "t2-percent": 0.875, "valid-lifetime": 3600}], "tl-percent": 0.5, "t2-percent": 0.875, "valid-lifetime": 3600}, "hash": "2E0668B1087DE8D555A48D2066AD59DE3299C79D7EC67F5899C8DBE888FAH"}, "result": 0}]
smartinez6@kea-server-shailamartinez:~$
```

Al Virtualbox al adaptador de xarxa de la màquina client SMG-UD1 (ubuntu desktop 26.04), li assigno de manera estàtica la MAC 08:00:27:11:22:33 per simular la màquina del servidor d'impressió.

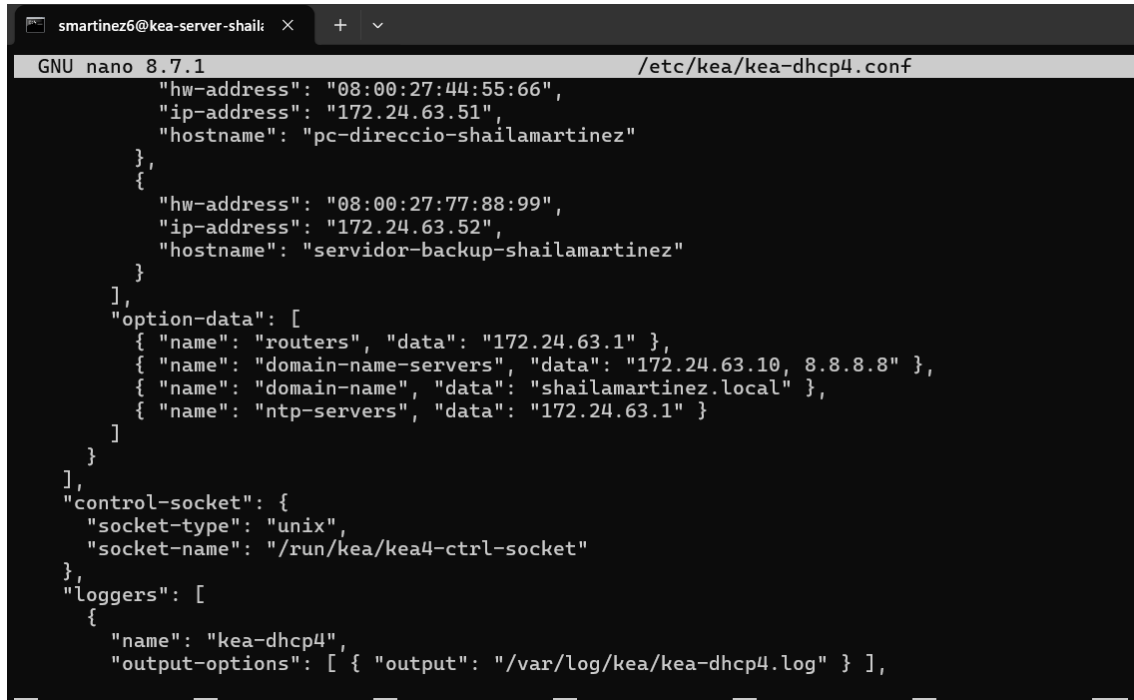


Primer s'allibera la concessió anterior i sollicita una nova, el client envia un paquet dhcpdiscover amb la MAC indicada a la captura anterior, el servidor Kea dona la IP reservada 172.24.63.50. Després comprovo l'assignació de la IP en el client és correcta.

```
smartinez6@SMG-UD1:~$ sudo dhclient -r enp0s3
smartinez6@SMG-UD1:~$ sudo dhclient enp0s3
Setting LLNMR support level "yes" for "2", but the global support level is "no".
smartinez6@SMG-UD1:~$ ip a show enp0s3
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:11:22:33 brd ff:ff:ff:ff:ff:ff
    altname enx080027112233
    inet 172.24.63.50/24 brd 172.24.63.255 scope global dynamic enp0s3
        valid_lft 3427sec preferred_lft 3427sec
    inet6 fe80::a00:27ff:fe11:2233/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
smartinez6@SMG-UD1:~$
```

Part 4: Opcions Addicionals i Consulta de Concessions

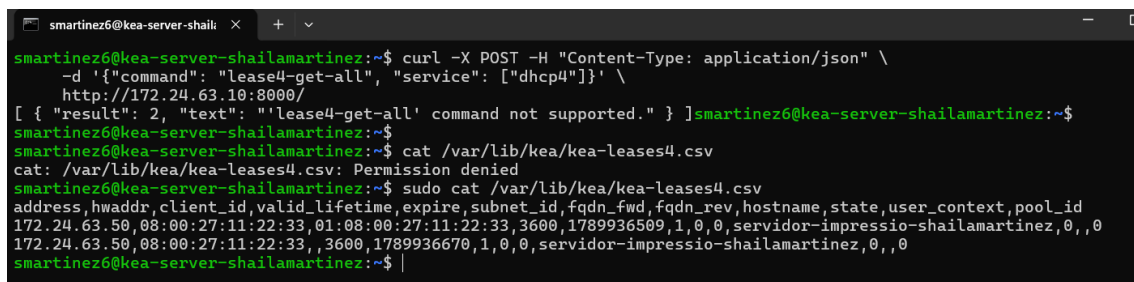
Al fitxer kea-dhcp4.conf indico la opció del server de temps (NTP) en “option-data” amb la IP i domini corresponent.



```
smartinez6@kea-server-shail: X + v
GNU nano 8.7.1 /etc/kea/kea-dhcp4.conf
    "hw-address": "08:00:27:44:55:66",
    "ip-address": "172.24.63.51",
    "hostname": "pc-direccio-shailamartinez"
  },
  {
    "hw-address": "08:00:27:77:88:99",
    "ip-address": "172.24.63.52",
    "hostname": "servidor-backup-shailamartinez"
  }
],
"option-data": [
  { "name": "routers", "data": "172.24.63.1" },
  { "name": "domain-name-servers", "data": "172.24.63.10, 8.8.8.8" },
  { "name": "domain-name", "data": "shailamartinez.local" },
  { "name": "ntp-servers", "data": "172.24.63.1" }
]
},
"control-socket": {
  "socket-type": "unix",
  "socket-name": "/run/kea/kea4-ctrl-socket"
},
"loggers": [
  {
    "name": "kea-dhcp4",
    "output-options": [ { "output": "/var/log/kea/kea-dhcp4.log" } ],

```

Es consulta les concessions i lectura de fitxer de leases (lease4-get-all), retorna un missatge de command not supported. Això és perquè lease4-get-all no forma part de les comandes bàsiques de Kea.



```
smartinez6@kea-server-shailamartinez:~$ curl -X POST -H "Content-Type: application/json" \
-d '{"command": "lease4-get-all", "service": ["dhcp4"]}' \
http://172.24.63.10:8000/
[ { "result": 2, "text": "'lease4-get-all' command not supported." } ]
smartinez6@kea-server-shailamartinez:~$
smartinez6@kea-server-shailamartinez:~$ cat /var/lib/kea/kea-leases4.csv
cat: /var/lib/kea/kea-leases4.csv: Permission denied
smartinez6@kea-server-shailamartinez:~$ sudo cat /var/lib/kea/kea-leases4.csv
address,hwaddr,client_id,valid_lifetime,expire,subnet_id,fqdn_fwd,fqdn_rev,hostname,state,user_context,pool_id
172.24.63.50,08:00:27:11:22:33,01:08:00:27:11:22:33,3600,1789936509,1,0,0,servidor-impressio-shailamartinez,0,,0
172.24.63.50,08:00:27:11:22:33,,3600,1789936670,1,0,0,servidor-impressio-shailamartinez,0,,0
smartinez6@kea-server-shailamartinez:~$ |
```

S’ha d’activar la llibreria de hooks de Kea en el servidor, sinó no reconeix la comanda per API. Afegeixo aquestes línies:

```
smartinez6@kea-server-shaili x + v
GNU nano 8.7.1 /etc/kea/kea-dhcp4.conf
{
  "Dhcp4": {
    "interfaces-config": {
      "interfaces": [ "enp0s3" ]
    },
    "lease-database": {
      "type": "memfile",
      "persist": true,
      "name": "/var/lib/kea/kea-leases4.csv"
    },
    "valid-lifetime": 3600,
    "renew-timer": 1800,
    "rebind-timer": 3150,
    "hooks-libraries": [
      {
        "library": "/usr/lib/x86_64-linux-gnu/kea/hooks/libdhcp_lease_cmds.so"
      }
    ],
    "subnet4": [
      {
        "id": 1,
        "subnet": "172.24.63.0/24",
        "pools": [ { "pool": "172.24.63.100 - 172.24.63.200" } ],
        "reservations": [ ]
      }
    ]
  }
}
```

Comanda amb la llibreria activada per consultar les concessions actives per Api:

```
smartinez6@kea-server-shailamartinez:~$ curl -X POST -H "Content-Type: application/json" \
-d '{"command": "lease4-get-all", "service": ["dhcp4"]}' \
http://172.24.63.10:8000/
[ { "arguments": { "leases": [ { "cltt": 1790022077, "fqdn-fwd": false, "fqdn-rev": false, "hostname": "servidor-impressio-shailamartinez", "hw-address": "08:00:27:11:22:33", "ip-address": "172.24.63.50", "state": 0, "subnet-id": 1, "valid-lft": 3600 } ] }, "result": 0, "text": "1 IPv4 lease(s) found." } ]
smartinez6@kea-server-shailamartinez:~$
```

Cat:

```
smartinez6@kea-server-shaili x + v
smartinez6@kea-server-shailamartinez:~$ sudo cat /var/lib/kea/kea-leases4.csv
address,hwaddr,client_id,valid_lifetime,expire,subnet_id,fqdn_fwd,fqdn_rev,hostname,state,user_context,pool_id
172.24.63.50,08:00:27:11:22:33,01:08:00:27:11:22:33,3600,1789938309,1,0,0,2,,0
172.24.63.50,08:00:27:11:22:33,01:08:00:27:11:22:33,0,1789934709,1,0,0,2,,0
172.24.63.50,08:00:27:11:22:33,01:08:00:27:11:22:33,0,1789934709,1,0,0,2,,0
172.24.63.50,08:00:27:11:22:33,01:08:00:27:11:22:33,3600,1790025357,1,0,0,servidor-impressio-shailamartinez,0,,0
172.24.63.50,08:00:27:11:22:33,01:08:00:27:11:22:33,0,1790021757,1,0,0,servidor-impressio-shailamartinez,3,,0
172.24.63.50,08:00:27:11:22:33,,3600,1790025677,1,0,0,servidor-impressio-shailamartinez,0,,0
smartinez6@kea-server-shailamartinez:~$ |
```

Taula concessions actives:

Ara només hi ha 1 concessió activa perquè només la màquina amb la MAC 08:00:27:11:22:33 ha iniciat la negociació DHCP. Les IPs 172.24.63.51 i 172.24.63.52 estan reservades al fitxer de configuració, no generaran una concessió activa fins que els equips es connectin a la xarxa i demanar una IP.

IP	MAC	Hostname	Temps venciment
172.24.63.50	08:00:27:11:22:33	servidor-impressio-shailamartinez	1790025677

Per comprovar, he fet una prova i he indicat 2 adaptadors més de xarxa per veure si rep la IP correctament (simulant 2 equips més)

Adaptador 1	Adaptador 2	Adaptador 3	Adaptador 4
☒ Habilita l'adaptador de xarxa			
Connectat a Xarxa interna			
Nom intnet			
Tipus d'adaptador Intel PRO/1000 MT Desktop (82540EM)			
Mode promiscu Denega			
Adreça MAC 080027445566			

Adaptador 1	Adaptador 2	Adaptador 3	Adaptador 4
☒ Habilita l'adaptador de xarxa			
Connectat a Xarxa interna			
Nom intnet			
Tipus d'adaptador Intel PRO/1000 MT Desktop (82540EM)			
Mode promiscu Denega			
Adreça MAC 080027778899			

Comprovació de les IPs: podem veure que reb la .50 (la primera que va rebre a la part 3), la .51 i .52

```
21 de sep 23:40
smartinez6@SMG-UD1: ~
smartinez6@SMG-UD1:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
   link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
   inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
   inet6 ::1/128 scope host noprefixroute
       valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
   link/ether 08:00:27:11:22:33 brd ff:ff:ff:ff:ff:ff
   altnam enx080027112233
   inet 172.24.63.50/24 brd 172.24.63.255 scope global dynamic noprefixroute enp0s3
       valid_lft 3565sec preferred_lft 3565sec
   inet6 fe80::a00:27ff:fe11:2233/64 scope link noprefixroute
       valid_lft forever preferred_lft forever
3: enp0s8: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
   link/ether 08:00:27:44:55:66 brd ff:ff:ff:ff:ff:ff
   altnam enx080027445566
   inet 172.24.63.51/24 brd 172.24.63.255 scope global dynamic noprefixroute enp0s8
       valid_lft 3566sec preferred_lft 3566sec
   inet6 fe80::9b01:c9b5:a7c0:bad7/64 scope link noprefixroute
       valid_lft forever preferred_lft forever
4: enp0s9: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
   link/ether 08:00:27:77:88:99 brd ff:ff:ff:ff:ff:ff
   altnam enx080027778899
   inet 172.24.63.52/24 brd 172.24.63.255 scope global dynamic enp0s9
       valid_lft 3567sec preferred_lft 3567sec
   inet6 fe80::f933:5cda:cc7c:82e6/64 scope link noprefixroute
       valid_lft forever preferred_lft forever
smartinez6@SMG-UD1:~$
```

Servidor:

```
smartinez6@kea-server-shaila:~$ curl -X POST -H "Content-Type: application/json" \
-d '{"command": "lease4-get-all", "service": ["dhcp4"]}' \
http://172.24.63.10:8000/
[ { "arguments": { "leases": [ { "cltt": 1790026809, "fqdn-fwd": false, "fqdn-rev": false, "hostname": "servidor-impressio-shailamartinez", "hw-address": "08:00:27:11:22:33", "ip-address": "172.24.63.50", "state": 0, "subnet-id": 1, "valid-lft": 3600 }, { "cltt": 1790026809, "fqdn-fwd": false, "fqdn-rev": false, "hostname": "pc-direccio-shailamartinez", "hw-address": "08:00:27:44:55:66", "ip-address": "172.24.63.51", "state": 0, "subnet-id": 1, "valid-lft": 3600 }, { "cltt": 1790026810, "fqdn-fwd": false, "fqdn-rev": false, "hostname": "servidor-backup-shailamartinez", "hw-address": "08:00:27:77:88:99", "ip-address": "172.24.63.52", "state": 0, "subnet-id": 1, "valid-lft": 3600 }, { "cltt": 1790026670, "fqdn-fwd": false, "fqdn-rev": false, "hostname": "smg-ud1", "hw-address": "08:00:27:77:88:00", "ip-address": "172.24.63.100", "state": 0, "subnet-id": 1, "valid-lft": 3600 } ] }, "result": 0, "text": "4 IPv4 lease(s) found." } ]
smartinez6@kea-server-shailamartinez:~$ sudo cat /var/lib/kea/kea-leases4.csv
[sudo: authenticate] Password:
address,hwaddr,client_id,valid_lifetime,expire,subnet_id,fqdn_fwd,fqdn_rev,hostname,state,user_context,pool_id
172.24.63.50,08:00:27:11:22:33,01:08:00:27:11:22:33,3600,1790030233,1,0,0,servidor-impressio-shailamartinez,0,,0
172.24.63.100,08:00:27:77:88:00,01:08:00:27:77:88:00,3600,1790030235,1,0,0,smg-ud1,0,,0
172.24.63.51,08:00:27:44:55:66,01:08:00:27:44:55:66,3600,1790030235,1,0,0,pc-direccio-shailamartinez,0,,0
172.24.63.51,08:00:27:44:55:66,,3600,1790030270,1,0,0,pc-direccio-shailamartinez,0,,0
172.24.63.100,08:00:27:77:88:00,,3600,1790030270,1,0,0,smg-ud1,0,,0
172.24.63.50,08:00:27:11:22:33,,3600,1790030270,1,0,0,servidor-impressio-shailamartinez,0,,0
172.24.63.50,08:00:27:11:22:33,01:08:00:27:11:22:33,3600,1790030376,1,0,0,servidor-impressio-shailamartinez,0,,0
172.24.63.52,08:00:27:77:88:99,01:08:00:27:77:88:99,3600,1790030377,1,0,0,servidor-backup-shailamartinez,0,,0
172.24.63.51,08:00:27:44:55:66,01:08:00:27:44:55:66,3600,1790030378,1,0,0,pc-direccio-shailamartinez,0,,0
172.24.63.51,08:00:27:44:55:66,,3600,1790030409,1,0,0,pc-direccio-shailamartinez,0,,0
172.24.63.50,08:00:27:11:22:33,,3600,1790030409,1,0,0,servidor-impressio-shailamartinez,0,,0
172.24.63.52,08:00:27:77:88:99,,3600,1790030410,1,0,0,servidor-backup-shailamartinez,0,,0
smartinez6@kea-server-shailamartinez:~$
```

La taula quedaria així:

IP	MAC	Hostname	Temps venciment
172.24.63.50	08:00:27:11:22:33	servidor-impressio-shailamartinez	1790026809
172.24.63.51	08:00:27:44:55:66	pc-direccio-shailamartinez	1790026809
172.24.63.52	08:00:27:77:88:99	servidor-backup-shailamartinez	1790026810
172.24.63.100	08:00:27:77:88:00	smg-ud1	1790026670

La .100 és la primera assignada dinàmica pel pool (100-200).

Qüestions

1. Per què ISC DHCP està en fi de vida (EOL) i quins avantatges tècnics aporta Kea (arquitectura per hooks, API REST, configuració JSON)?

El ISC DHCP va deixar de desenvolupar-se a 2022 i Kea seria el successor pel mateix fabricant ISC. Avantatges principals:

Es configura en JSON, té una API REST per gestionar i consultar el servei, integra alta disponibilitat de forma nativa, suporta bases de dades com MySQL i Postgresql, arquitectura per hooks que permet afegir funcionalitats mitjançant mòduls externs sense haver de canviar el codi principal.

2. Quina diferència hi ha entre una reserva (reservations) i el pool dinàmic (pools)? Quan s'ha d'utilitzar cadascuna?

El pool dinàmic és un rang de IPs que el DHCP assigna automàticament i de forma temporal als clients, quan el lease time expira, si l'equip no l'ha renovat, la IP torna al pool i pot ser assignada a un altre equip. Es fan servir per a dispositius comuns que no necessiten una IP fixa, com ordinadors, mòbils, xarxes de convidats.

En canvi una reserva assignen (de forma fixa) una IP concreta a una MAC determinada i l'equip sempre rebrà la mateixa IP. S'utilitzen per a servidors, impressores, routers (han de ser localitzables).

3. Per a què serveix kea-ctrl-agent i per què és útil exposar-lo via API REST en lloc de només fitxers de configuració?

kea-ctrl-agent rep ordres externes mitjançant una API REST i les comunica al servidor Kea DHCP perquè les executi.

És útil perquè permet consultar i gestionar el servei de forma remota i automatitzada, sense haver d'accedir directament als fitxers de configuració. També facilita la integració amb altres programes i eines d'administració.

4. Quins riscos de seguretat implica un servidor DHCP rogue a la xarxa, i com es podria mitigar (relació amb DHCP Snooping)?

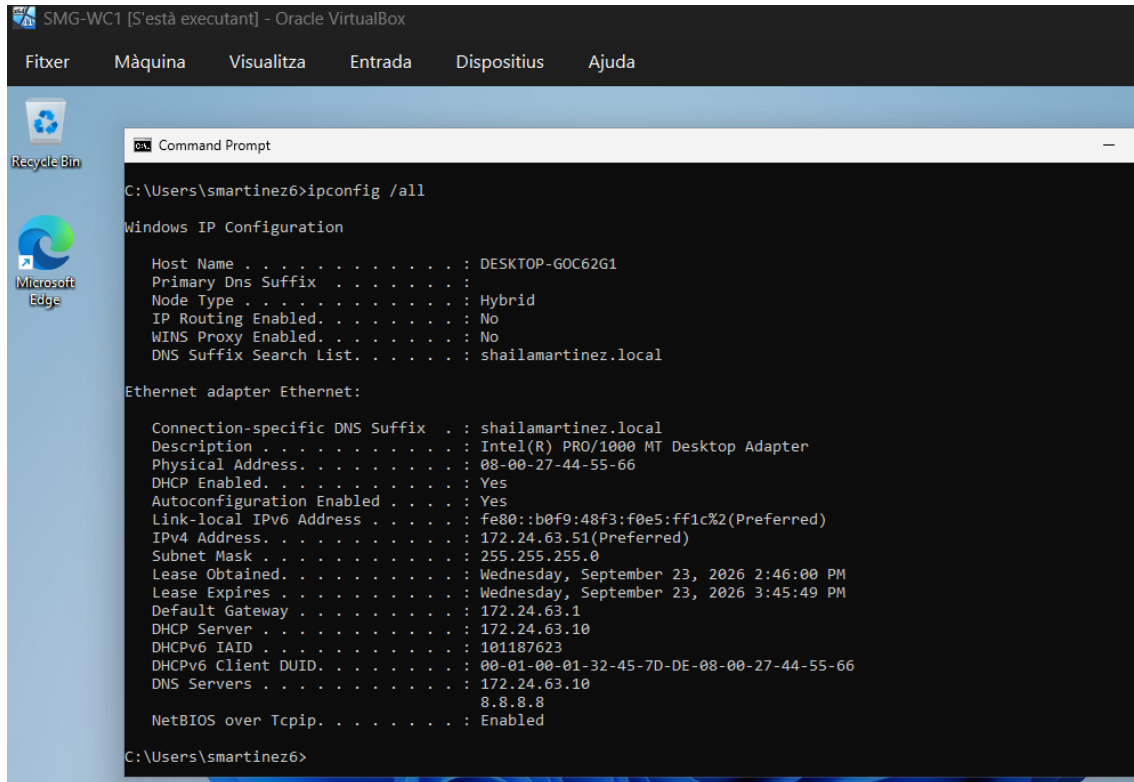
Un DHCP rogue és un servidor no autoritzat que pot proporcionar configuracions falses als clients de la xarxa que pot provocar problemes de connexió o que el tràfic sigui redirigit.

Una mesura de protecció amb DHCP Snooping permet indicar els ports del switch són de confiança. Si el switch detecta respostes DHCP que venen d'un port no autoritzat, pot bloquejar aquest tràfic.

Exercicis Moodle

- a) Obteniu l'adreça MAC amb la qual esteu fent les proves

Al client Windows 11 li he assignat la MAC 08:00:27:44:55:66 perquè obtingui la IP 171.24.63.51



```
SMG-WC1 [S'està executant] - Oracle VirtualBox
Fitxer  Màquina  Visualitza  Entrada  Dispositius  Ajuda

Recycle Bin
Microsoft Edge

C:\Users\smartinez6>ipconfig /all

Windows IP Configuration

Host Name . . . . . : DESKTOP-G0C62G1
Primary Dns Suffix . . . . . :
Node Type . . . . . : Hybrid
IP Routing Enabled. . . . . : No
WINS Proxy Enabled. . . . . : No
DNS Suffix Search List. . . . . : shailamartinez.local

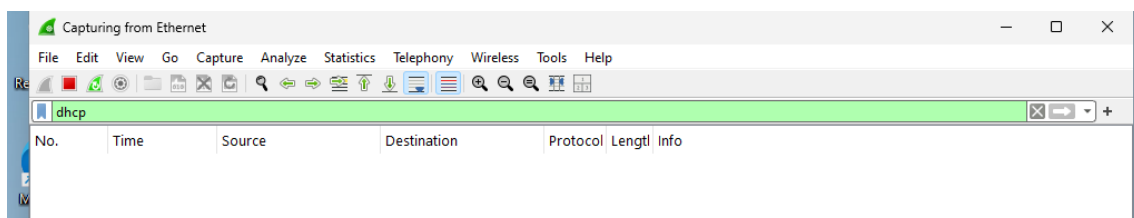
Ethernet adapter Ethernet:

Connection-specific DNS Suffix . : shailamartinez.local
Description . . . . . : Intel(R) PRO/1000 MT Desktop Adapter
Physical Address. . . . . : 08-00-27-44-55-66
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . : Yes
Link-local IPv6 Address . . . . . : fe80::b0f9:48f3:f0e5:ff1c%2(Preferred)
IPv4 Address. . . . . : 172.24.63.51(Preferred)
Subnet Mask . . . . . : 255.255.255.0
Lease Obtained. . . . . : Wednesday, September 23, 2026 2:46:00 PM
Lease Expires . . . . . : Wednesday, September 23, 2026 3:45:49 PM
Default Gateway . . . . . : 172.24.63.1
DHCP Server . . . . . : 172.24.63.10
DHCPv6 IAID . . . . . : 101187623
DHCPv6 Client DUID. . . . . : 00-01-00-01-32-45-7D-08-00-27-44-55-66
DNS Servers . . . . . : 172.24.63.10
                        8.8.8.8
NetBIOS over Tcpip. . . . . : Enabled

C:\Users\smartinez6>
```

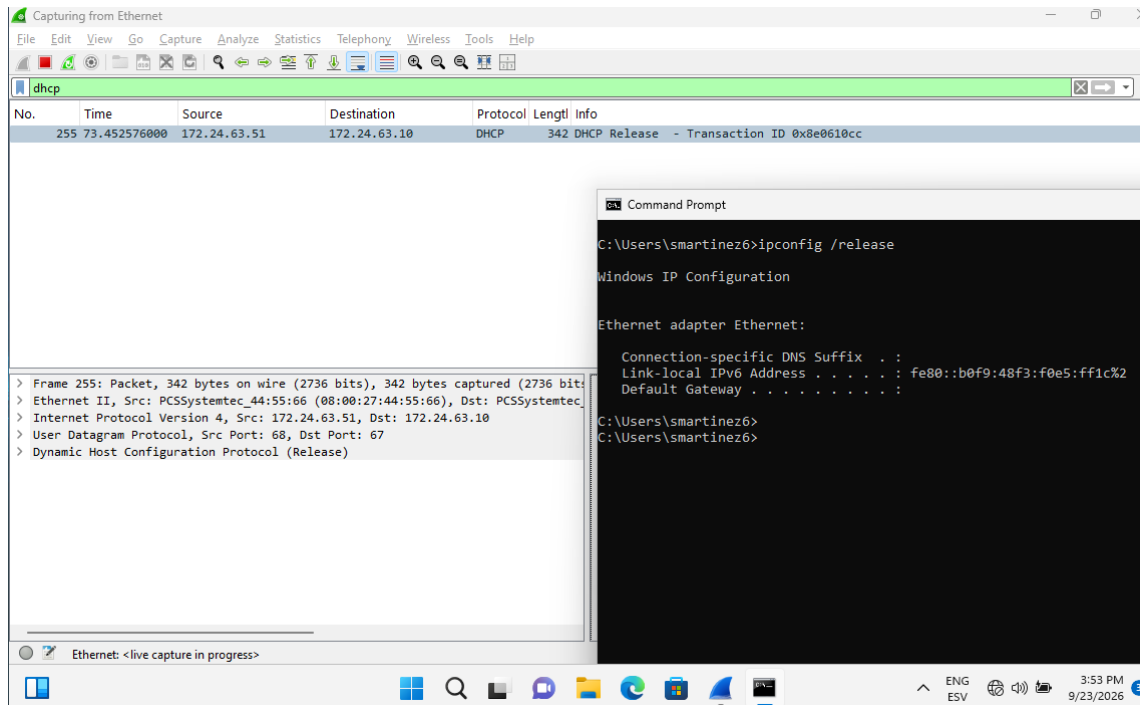
- b) Aplicar un filtre perquè només us mostri el trànsit corresponent al protocol DHCP

Filtre de DHCP:



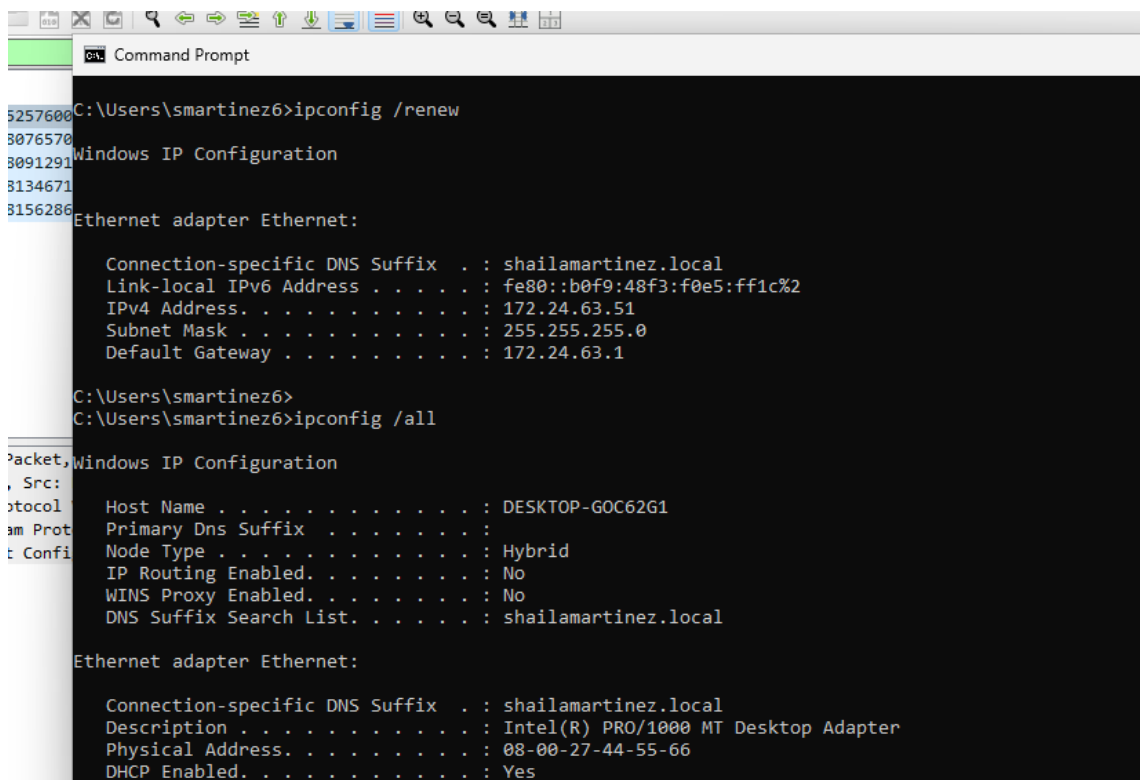
- c) Captura de la comanda de Windows conforme heu llançat la petició per alliberar la vostra NIC i que es quedi sense IP

Petició sense IP: `ipconfig /release`



- d) Forceu el client de DHCP de Windows per tal que solliciti una nova IP.
Adjunteu la captura on es veu la comanda de Windows que utilitzeu i demostreu que us ha assignat una IP vinculada a la MAC que heu indicat als apartats anteriors

Sollicitud de la IP nova amb renew:



- e) Caldrà que adjunteu les 4 captures del Wireshark, una per cadascuna de les fases bàsiques del protocol. Wireshark treballa amb una finestra dividida en 3 zones. Per cada captura caldrà que es mostrin la zona superior i la del mig. La del mig interessa perquè hi veureu les MACs i els ports i demostrareu que la petició es vostra. 2 punts.

Cicle de negociació amb els paquets corresponents:

1:

Frame 307: Packet, 344 bytes on wire (2752 bits), 344 bytes captured (2752 bits) on interface 0

Ethernet II, Src: PCSSystemtec_44:55:66 (08:00:27:44:55:66), Dst: Broadcast (ff:ff:ff:ff:ff:ff)

Destination: Broadcast (ff:ff:ff:ff:ff:ff)

Source: PCSSystemtec_44:55:66 (08:00:27:44:55:66)

Type: IPv4 (0x0800)

[Stream index: 0]

Internet Protocol Version 4, Src: 0.0.0.0, Dst: 255.255.255.255

User Datagram Protocol, Src Port: 68, Dst Port: 67

Source Port: 68

Destination Port: 67

Length: 310

Checksum: 0xa588 [unverified]

[Checksum Status: Unverified]

[Stream index: 97]

[Stream Packet Number: 1]

[Timestamps]

UDP payload (302 bytes)

Dynamic Host Configuration Protocol (Discover)

2:

Frame 308: Packet, 391 bytes on wire (3128 bits), 391 bytes captured (3128 bits) on interface 0

Ethernet II, Src: PCSSystemtec_d4:73:fa (08:00:27:d4:73:fa), Dst: PCSSystemtec_44:55:66 (08:00:27:44:55:66)

Destination: PCSSystemtec_44:55:66 (08:00:27:44:55:66)

Source: PCSSystemtec_d4:73:fa (08:00:27:d4:73:fa)

Type: IPv4 (0x0800)

[Stream index: 1]

Internet Protocol Version 4, Src: 172.24.63.10, Dst: 172.24.63.10

User Datagram Protocol, Src Port: 67, Dst Port: 68

Source Port: 67

Destination Port: 68

Length: 357

Checksum: 0xe4d9 [unverified]

[Checksum Status: Unverified]

[Stream index: 90]

[Stream Packet Number: 2]

[Timestamps]

UDP payload (349 bytes)

Dynamic Host Configuration Protocol (Offer)

3:

Capturing from Ethernet

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

dhcpc

No.	Time	Source	Destination	Protocol	Length	Info
255	73.452576000	172.24.63.51	172.24.63.10	DHCP	342	DHCP Release - Transaction ID 0x8e0610cc
307	160.807657000	0.0.0.0	255.255.255.255	DHCP	344	DHCP Discover - Transaction ID 0x43729307
308	160.809129100	172.24.63.10	172.24.63.51	DHCP	391	DHCP Offer - Transaction ID 0x43729307
309	160.813467100	0.0.0.0	255.255.255.255	DHCP	370	DHCP Request - Transaction ID 0x43729307
310	160.815628600	172.24.63.10	172.24.63.51	DHCP	395	DHCP ACK - Transaction ID 0x43729307

> Frame 309: Packet, 370 bytes on wire (2960 bits), 370 bytes captured (2960 bits) on interface 0

> Ethernet II, Src: PCSSystemtec_44:55:66 (08:00:27:44:55:66), Dst: Broadcast (ff:ff:ff:ff:ff:ff)

> Destination: Broadcast (ff:ff:ff:ff:ff:ff)

> Source: PCSSystemtec_44:55:66 (08:00:27:44:55:66)

Type: IPv4 (0x0800)

[Stream index: 0]

> Internet Protocol Version 4, Src: 0.0.0.0, Dst: 255.255.255.255

> User Datagram Protocol, Src Port: 68, Dst Port: 67

Source Port: 68

Destination Port: 67

Length: 336

Checksum: 0x4602 [unverified]

[Checksum Status: Unverified]

[Stream index: 97]

[Stream Packet Number: 2]

[Timestamps]

UDP payload (328 bytes)

Dynamic Host Configuration Protocol (Request)

Ethernet: <live capture in progress> | Packets: 958 - Displayed: 5 (0.5%) | Profile: Default

4:

Capturing from Ethernet

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

dhcpc

No.	Time	Source	Destination	Protocol	Length	Info
255	73.452576000	172.24.63.51	172.24.63.10	DHCP	342	DHCP Release - Transaction ID 0x8e0610cc
307	160.807657000	0.0.0.0	255.255.255.255	DHCP	344	DHCP Discover - Transaction ID 0x43729307
308	160.809129100	172.24.63.10	172.24.63.51	DHCP	391	DHCP Offer - Transaction ID 0x43729307
309	160.813467100	0.0.0.0	255.255.255.255	DHCP	370	DHCP Request - Transaction ID 0x43729307
310	160.815628600	172.24.63.10	172.24.63.51	DHCP	395	DHCP ACK - Transaction ID 0x43729307

> Frame 310: Packet, 395 bytes on wire (3160 bits), 395 bytes captured (3160 bits) on interface 0

> Ethernet II, Src: PCSSystemtec_d4:73:fa (08:00:27:d4:73:fa), Dst: PCSSystemtec_d4:73:fa (08:00:27:d4:73:fa)

> Destination: PCSSystemtec_d4:73:fa (08:00:27:d4:73:fa)

> Source: PCSSystemtec_d4:73:fa (08:00:27:d4:73:fa)

Type: IPv4 (0x0800)

[Stream index: 1]

> Internet Protocol Version 4, Src: 172.24.63.10, Dst: 172.24.63.51

> User Datagram Protocol, Src Port: 67, Dst Port: 68

Source Port: 67

Destination Port: 68

Length: 361

Checksum: 0xa291 [unverified]

[Checksum Status: Unverified]

[Stream index: 90]

[Stream Packet Number: 3]

[Timestamps]

UDP payload (353 bytes)

Dynamic Host Configuration Protocol (ACK)

Ethernet: <live capture in progress> | Packets: 990 - Displayed: 5 (0.5%) | Profile: Default

- f) Ara feu una petició de renovació d'IP des de la consola de Windows. No cal que indiqueu la comanda. Com que encara no haurà caducat el temps de concessió, la captura al Wireshark serà diferent. Adjunteu una captura on es mostri la finestra superior del programa.

Intercanvi de renovació 2 paquets:

```

C:\Users\smartinez6>ipconfig /renew

Windows IP Configuration

Ethernet adapter Ethernet:

    Connection-specific DNS Suffix  . : shailamartinez.local
    Link-local IPv6 Address . . . . . : fe80::b0f9:48f3:f0e5:ff1c%2
    IPv4 Address. . . . . : 172.24.63.51
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : 172.24.63.1

C:\Users\smartinez6>
C:\Users\smartinez6>
C:\Users\smartinez6>
```

DHCP Request:

Capturing from Ethernet

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

dhcp

No.	Time	Source	Destination	Protocol	Length	Info
179	184.822411200	172.24.63.51	172.24.63.10	DHCP	358	DHCP Request - Transaction ID 0x7c204a90
180	184.824035200	172.24.63.10	172.24.63.51	DHCP	395	DHCP ACK - Transaction ID 0x7c204a90

> Frame 179: Packet, 358 bytes on wire (2864 bits), 358 bytes captured (2864 bits) on interface 0

Ethernet II, Src: PCSSystemtec_44:55:66 (08:00:27:44:55:66), Dst: PCSSystemtec_44:55:66 (08:00:27:44:55:66)

> Destination: PCSSystemtec_44:55:66 (08:00:27:44:55:66)

> Source: PCSSystemtec_44:55:66 (08:00:27:44:55:66)

Type: IPv4 (0x0800)

[Stream index: 1]

> Internet Protocol Version 4, Src: 172.24.63.51, Dst: 172.24.63.10

> User Datagram Protocol, Src Port: 68, Dst Port: 67

Source Port: 68

Destination Port: 67

Length: 324

Checksum: 0xd7c3 [unverified]

[Checksum Status: Unverified]

[Stream index: 29]

[Stream Packet Number: 1]

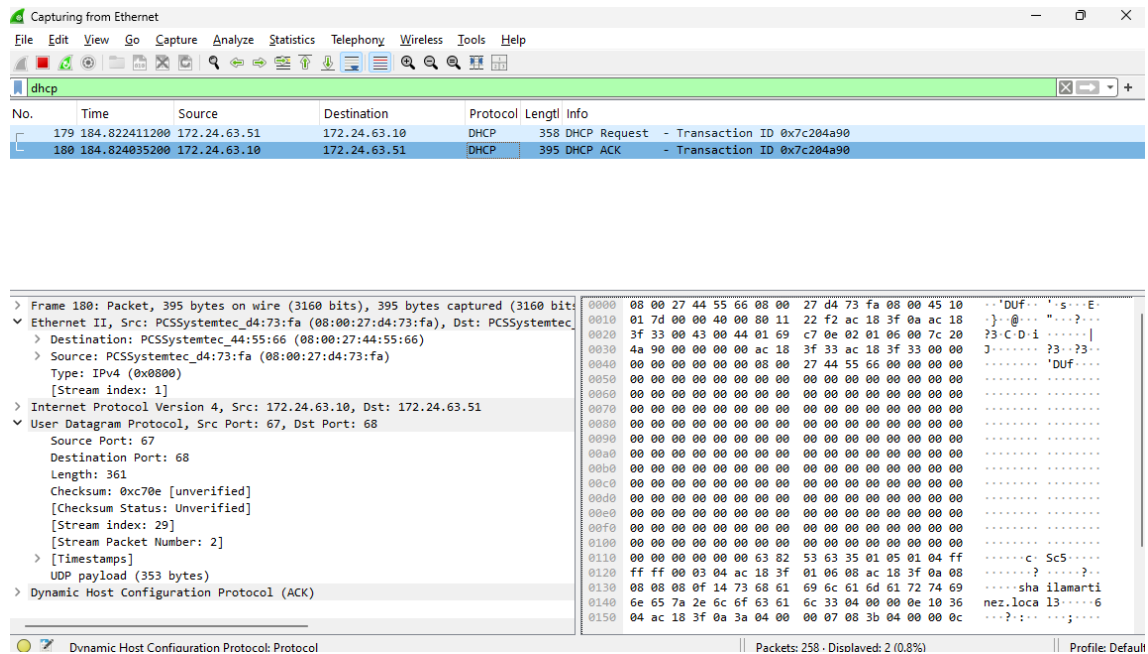
[Timestamps]

UDP payload (316 bytes)

Dynamic Host Configuration Protocol (Request)

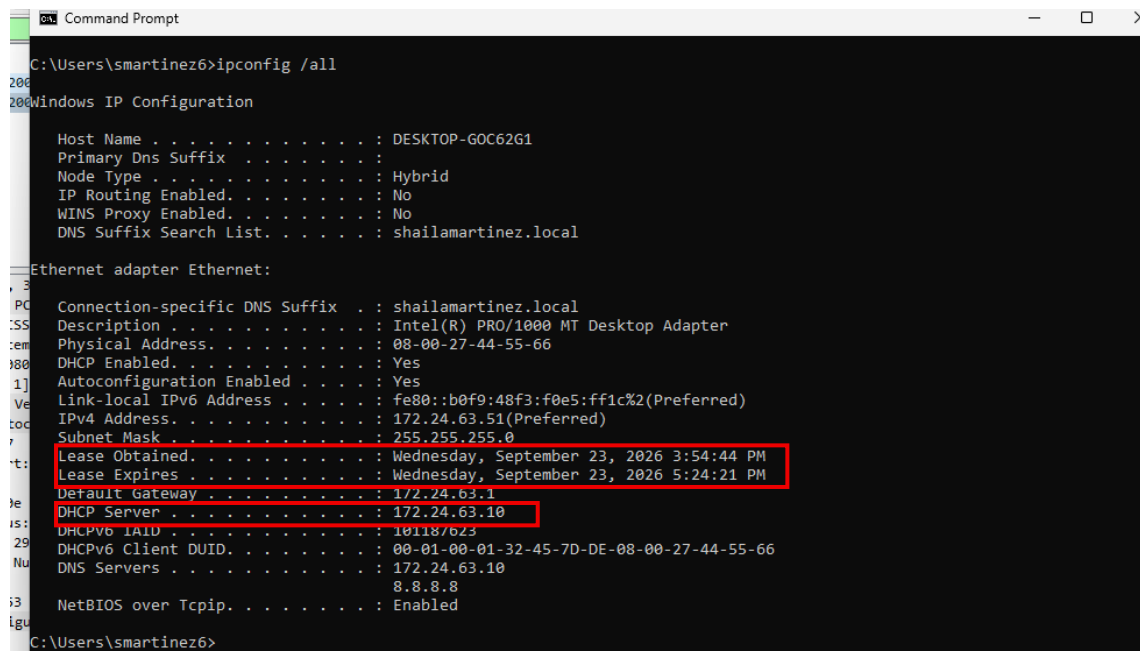
0000	08 00 27 d4 73 fa 08 00 27 44 55 66 08 00 45 00	s...DUF..E..
0010	01 58 42 db 00 00 00 11 00 00 ac 18 3f 33 ac 18	..XB.....?3...
0020	3f 0a 00 44 00 43 01 44 d7 c3 01 01 06 00 7c 20	...D.C.D.....
0030	4a 90 00 00 00 00 ac 18 3f 33 00 00 00 00 00 00	J.....?3.....
0040	00 00 00 00 00 00 00 00 27 44 55 66 00 00 00 00	DUF.....
0050	00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00	
0060	00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00	
0070	00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00	
0080	00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00	
0090	00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00	
00a0	00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00	
00b0	00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00	
00c0	00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00	
00d0	00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00	
00e0	00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00	
00f0	00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00	
0100	00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00	
0110	00 00 00 00 00 00 63 82 53 63 35 01 03 3d 07 01	c..ScS...=...
0120	08 00 27 44 55 66 0c 0f 44 45 53 4b 54 4f 50 2d	...DUF...DESKTOP-
0130	47 4f 43 36 32 47 31 51 12 00 00 00 44 45 53 4b	GOC6261Q.....DESK
0140	54 4f 50 2d 47 4f 43 36 32 47 31 3c 08 4d 53 4b	TOP-GOC6 261<MSF
0150	54 20 35 2e 30 37 0e 01 03 06 0f 1f 21 2b 2c 2e	T 5.07.....!+,..

DHCP ACK:



- g) Finalment, amb la comanda `ipconfig`, passeu una captura on es vegi el temps de la concessió (quan es va obtenir i quan expira) ni quina és la IP del servidor DHCP. Remarqueu aquests 3 paràmetres dins la captura

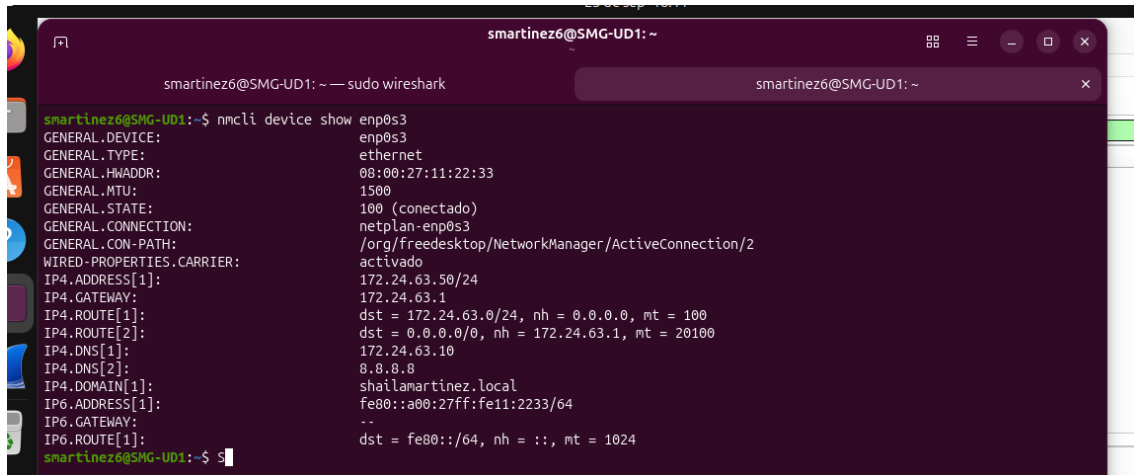
Lease obtained, Lease expires i la IP del server DHCP:



- h) Per tant, un cop hagi agafat IP, utilitzeu la comanda `nmcli` per veure la informació obtinguda per la vostra connexió via DHCP. Cal que mostri els

paràmetres obtinguts específicament per la vostra connexió (IP, màscara, gateway, DNS, etc.). Adjunteu una captura que ho demostrï.

Al client Ubuntu amb nmcli consulta els paràmetres de la xarxa.



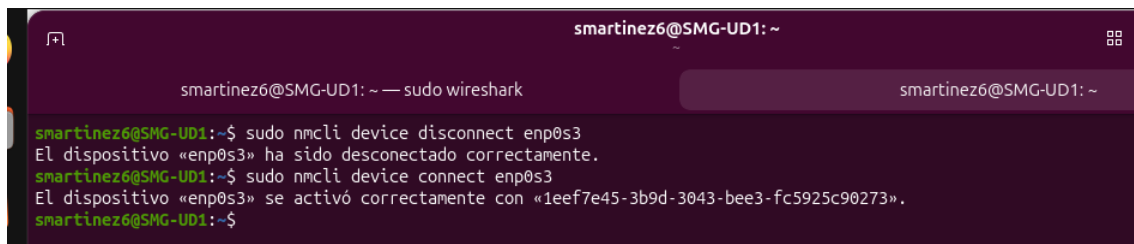
```
smartinez6@SMG-UD1: ~ - sudo wireshark
smartinez6@SMG-UD1: ~$ nmcli device show enp0s3
GENERAL.DEVICE: enp0s3
GENERAL.TYPE: ethernet
GENERAL.HWADDR: 08:00:27:11:22:33
GENERAL.MTU: 1500
GENERAL.STATE: 100 (conectado)
GENERAL.CONNECTION: netplan-enp0s3
GENERAL.CON-PATH: /org/freedesktop/NetworkManager/ActiveConnection/2
WIRED-PROPERTIES.CARRIER: activado
IP4.ADDRESS[1]: 172.24.63.50/24
IP4.GATEWAY: 172.24.63.1
IP4.ROUTE[1]: dst = 172.24.63.0/24, nh = 0.0.0.0, mt = 100
IP4.ROUTE[2]: dst = 0.0.0.0/0, nh = 172.24.63.1, mt = 20100
IP4.DNS[1]: 172.24.63.10
IP4.DNS[2]: 8.8.8.8
IP4.DOMAIN[1]: shailamartinez.local
IP6.ADDRESS[1]: fe80::a00:27ff:fe11:2233/64
IP6.GATEWAY: --
IP6.ROUTE[1]: dst = fe80::/64, nh = ::, mt = 1024
smartinez6@SMG-UD1: ~$
```

- i) Busqueu com fer amb nmcli per tal d'alliberar i sol·licitar novament la IP. Adjunteu novament les captures del procés que ho demostrin.

`sudo nmcli device disconnect enp0s3`

`sudo nmcli device connect enp0s3`

Allibera la IP i la reactiva:



```
smartinez6@SMG-UD1: ~
smartinez6@SMG-UD1: ~ - sudo wireshark
smartinez6@SMG-UD1: ~$ sudo nmcli device disconnect enp0s3
El dispositivo «enp0s3» ha sido desconectado correctamente.
smartinez6@SMG-UD1: ~$ sudo nmcli device connect enp0s3
El dispositivo «enp0s3» se activó correctamente con «1eef7e45-3b9d-3043-bee3-fc5925c90273».
smartinez6@SMG-UD1: ~$
```

- j) Caldrà adjuntar les N captures del Wireshark que demostrin tot el procés (s'ha de veure com s'allibera la IP i després es fa la petició des de 0). Les captures han de ser del mateix estil que l'apartat e).

Captures de DORA:

1:

SMG-UC1 [S'està executant] - Oracle VirtualBox

23 de sep 18:13

Capturing from enp0s3

dhcph

No.	Time	Source	Destination	Protocol	Length	Info
239	128.477661329	0.0.0.0	255.255.255.255	DHCP	333	DHCP Request - Transaction ID 0x46a04767
240	128.480653550	172.24.63.10	172.24.63.50	DHCP	398	DHCP ACK - Transaction ID 0x46a04767

Frame 240: Packet, 398 bytes on wire (3184 bits), 398 bytes captured (3184 bits) on enp0s3

Ethernet II, Src: PCSSTestec_d4:73:fa (08:00:27:d4:73:fa), Dst: PCSSTestec_11:22:33 (08:00:27:11:22:33)

Destination: Broadcast (ff:ff:ff:ff:ff:ff)

Source: PCSSTestec_11:22:33 (08:00:27:11:22:33)

Type: IPv4 (0x0800)

[Stream index: 1]

Internet Protocol Version 4, Src: 172.24.63.10, Dst: 172.24.63.50

User Datagram Protocol, Src Port: 67, Dst Port: 68

Source Port: 67

Destination Port: 68

Length: 364

Checksum: 0xe7da [unverified]

[Checksum Status: Unverified]

[Stream index: 37]

[Stream Packet Number: 1]

[Timestamps]

UDP payload (356 bytes)

Dynamic Host Configuration Protocol (ACK)

Bytes 287-290: Subnet Mask (dhcp.option.subnet_mask)

Paquetes: 409 - Displayed: 2 (0.5%)

Perfil: Default

2:

SMG-UC1 [S'està executant] - Oracle VirtualBox

23 de sep 18:13

Capturing from enp0s3

dhcph

No.	Time	Source	Destination	Protocol	Length	Info
239	128.477661329	0.0.0.0	255.255.255.255	DHCP	333	DHCP Request - Transaction ID 0x46a04767
240	128.480653550	172.24.63.10	172.24.63.50	DHCP	398	DHCP ACK - Transaction ID 0x46a04767

Frame 240: Packet, 398 bytes on wire (3184 bits), 398 bytes captured (3184 bits) on enp0s3

Ethernet II, Src: PCSSTestec_d4:73:fa (08:00:27:d4:73:fa), Dst: PCSSTestec_11:22:33 (08:00:27:11:22:33)

Destination: Broadcast (ff:ff:ff:ff:ff:ff)

Source: PCSSTestec_d4:73:fa (08:00:27:d4:73:fa)

Type: IPv4 (0x0800)

[Stream index: 0]

Internet Protocol Version 4, Src: 172.24.63.10, Dst: 172.24.63.50

User Datagram Protocol, Src Port: 67, Dst Port: 68

Source Port: 67

Destination Port: 68

Length: 364

Checksum: 0xe7da [unverified]

[Checksum Status: Unverified]

[Stream index: 37]

[Stream Packet Number: 1]

[Timestamps]

UDP payload (356 bytes)

Dynamic Host Configuration Protocol (ACK)

Bytes 287-290: Subnet Mask (dhcp.option.subnet_mask)

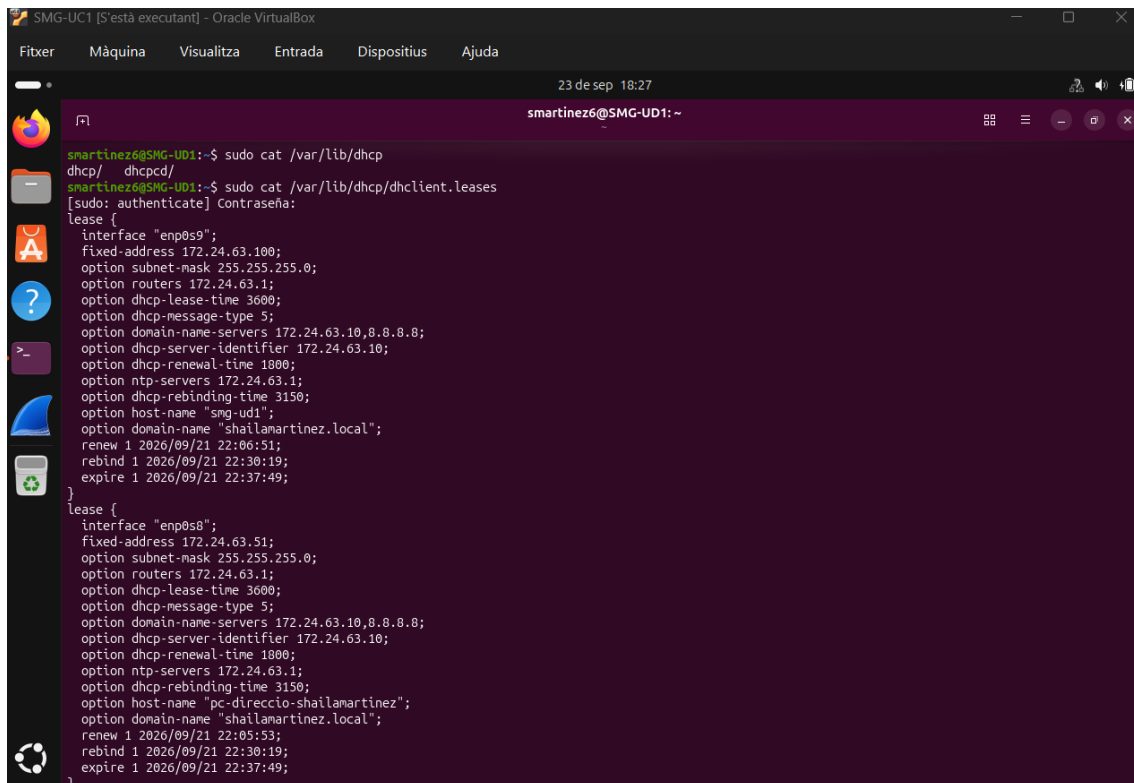
Paquetes: 455 - Displayed: 2 (0.4%)

Perfil: Default

- k) Tot seguit, cal que, dins la Ubuntu, busqueu el fitxer on hi ha la informació de 'lease' per la IP que teniu a la Ubuntu i que adjunteu una captura on es vegi la ruta del fitxer, i els paràmetres de 'lease' que us ha donat el servidor de DHCP.

Fitxer on es guarda la informació de les concessions:

`sudo cat /var/lib/dhcp/dhclient.leases`



```
SMG-UC1 [S'està executant] - Oracle VM VirtualBox
Fitxer  Màquina  Visualitza  Entrada  Dispositius  Ajuda
23 de sep 18:27
smartinez6@SMG-UD1: ~
smartinez6@SMG-UD1:~$ sudo cat /var/lib/dhcp
dhcp/  dhcpd/
smartinez6@SMG-UD1:~$ sudo cat /var/lib/dhcp/dhclient.leases
[sudo: authenticate] Contraseña:
lease {
  interface "enp0s9";
  fixed-address 172.24.63.100;
  option subnet-mask 255.255.255.0;
  option routers 172.24.63.1;
  option dhcp-lease-time 3600;
  option dhcp-message-type 5;
  option domain-name-servers 172.24.63.10,8.8.8.8;
  option dhcp-server-identifier 172.24.63.10;
  option dhcp-renewal-time 1800;
  option ntp-servers 172.24.63.1;
  option dhcp-rebinding-time 3150;
  option host-name "smg-ud1";
  option domain-name "shailamartinez.local";
  renew 1 2026/09/21 22:06:51;
  rebind 1 2026/09/21 22:30:19;
  expire 1 2026/09/21 22:37:49;
}
lease {
  interface "enp0s0";
  fixed-address 172.24.63.51;
  option subnet-mask 255.255.255.0;
  option routers 172.24.63.1;
  option dhcp-lease-time 3600;
  option dhcp-message-type 5;
  option domain-name-servers 172.24.63.10,8.8.8.8;
  option dhcp-server-identifier 172.24.63.10;
  option dhcp-renewal-time 1800;
  option ntp-servers 172.24.63.1;
  option dhcp-rebinding-time 3150;
  option host-name "pc-direccio-shailamartinez";
  option domain-name "shailamartinez.local";
  renew 1 2026/09/21 22:05:53;
  rebind 1 2026/09/21 22:30:19;
  expire 1 2026/09/21 22:37:49;
}
```

```
EXP 1 2026/09/21 22:31:42;
}
lease {
  interface "enp0s3";
  fixed-address 172.24.63.50;
  option subnet-mask 255.255.255.0;
  option routers 172.24.63.1;
  option dhcp-lease-time 3600;
  option dhcp-message-type 5;
  option domain-name-servers 172.24.63.10,8.8.8.8;
  option dhcp-server-identifier 172.24.63.10;
  option dhcp-renewal-time 1800;
  option ntp-servers 172.24.63.1;
  option dhcp-rebinding-time 3150;
  option host-name "servidor-impressio-shailamartinez";
  option domain-name "shailamartinez.local";
  renew 1 2026/09/21 22:02:44;
  rebind 1 2026/09/21 22:30:19;
  expire 1 2026/09/21 22:37:49;
}
lease {
  interface "enp0s8";
  fixed-address 172.24.63.51;
  option subnet-mask 255.255.255.0;
  option dhcp-lease-time 3600;
  option routers 172.24.63.1;
  option dhcp-message-type 5;
  option dhcp-server-identifier 172.24.63.10;
  option domain-name-servers 172.24.63.10,8.8.8.8;
  option dhcp-renewal-time 1800;
  option dhcp-rebinding-time 3150;
  option ntp-servers 172.24.63.1;
  option host-name "pc-direccio-shailamartinez";
  option domain-name "shailamartinez.local";
  renew 1 2026/09/21 22:03:14;
  rebind 1 2026/09/21 22:32:39;
  expire 1 2026/09/21 22:40:09;
}
lease {
  interface "enp0s3";
  fixed-address 172.24.63.50;
  expire 1 2026/09/21 22:40:09;
}
lease {
  interface "enp0s3";
  fixed-address 172.24.63.50;
  option subnet-mask 255.255.255.0;
  option dhcp-lease-time 3600;
  option routers 172.24.63.1;
  option dhcp-message-type 5;
  option dhcp-server-identifier 172.24.63.10;
  option domain-name-servers 172.24.63.10,8.8.8.8;
  option dhcp-renewal-time 1800;
  option dhcp-rebinding-time 3150;
  option ntp-servers 172.24.63.1;
  option host-name "servidor-impressio-shailamartinez";
  option domain-name "shailamartinez.local";
  renew 1 2026/09/21 22:07:11;
  rebind 1 2026/09/21 22:32:40;
  expire 1 2026/09/21 22:40:10;
}
lease {
  interface "enp0s9";
  fixed-address 172.24.63.52;
  option subnet-mask 255.255.255.0;
  option dhcp-lease-time 3600;
  option routers 172.24.63.1;
  option dhcp-message-type 5;
  option dhcp-server-identifier 172.24.63.10;
  option domain-name-servers 172.24.63.10,8.8.8.8;
  option dhcp-renewal-time 1800;
  option dhcp-rebinding-time 3150;
  option ntp-servers 172.24.63.1;
  option host-name "servidor-backup-shailamartinez";
  option domain-name "shailamartinez.local";
  renew 1 2026/09/21 22:05:41;
  rebind 1 2026/09/21 22:32:40;
  expire 1 2026/09/21 22:40:10;
}
smartinez6@SMG-UD1:~$
```

- l) Finalment, aneu a la darrera petició de 'DHCP ACK' dins el Wireshark i mostreu una captura amb els valors que s'han proporcionat Com a 'Options' dins la finestra del mig, i cliqueu sobre la opció 'Dynamic Host Configuration Protocol'.

Al últim paquet DHCP ACK es mostra el tipus de missatge, la màscara, gateway, nom del domini del server, IP lease time i identificador del server.

23 de sep 18:34

Capturing from enp0s3

Archivo Edición Visualización Ir Captura Analizar Estadísticas Telefonía Wireless Herramientas Ayuda

dhcp

No.	Time	Source	Destination	Protocol	Length	Info
166	137.459889199	0.0.0.0	255.255.255.255	DHCP	333	DHCP Request - Transaction ID 0x8dd98097
167	137.461845225	172.24.63.10	172.24.63.50	DHCP	404	DHCP ACK - Transaction ID 0x8dd98097

Next server IP address: 0.0.0.0
Relay agent IP address: 0.0.0.0
Client MAC address: PCSSystemtec_11:22:33 (08:00:27:11:22:33)
Client hardware address padding: 00000000000000000000
Server host name not given
Boot file name not given
Magic cookie: DHCP
Option: (53) DHCP Message Type (ACK)
Length: 1
DHCP: ACK (5)
Option: (1) Subnet Mask (255.255.255.0)
Length: 4
Subnet Mask: 255.255.255.0
Option: (3) Router
Length: 4
Router: 172.24.63.1
Option: (6) Domain Name Server
Length: 8
Domain Name Server: 172.24.63.10
Domain Name Server: 8.8.8.8
Option: (12) Host Name
Option: (15) Domain Name
Option: (42) Network Time Protocol Servers
Option: (51) IP Address Lease Time
Length: 4
IP Address Lease Time: 1 hour (3600)
Option: (54) DHCP Server Identifier (172.24.63.10)
Option: (58) Renewal Time Value
Option: (59) Rebinding Time Value

0000 08 00 27 11 22 33 08 00 27 d4 73 fa 08 00 45 10 ... "3..
0010 01 86 00 00 40 00 00 11 22 ea ac 18 3f 0a ac 18 ... @...
0020 3f 32 00 43 00 44 01 72 1d 19 02 01 06 00 8d d9 ?2 C.D.r
0030 80 97 00 00 00 00 00 00 00 00 ac 18 3f 32 00 00
0040 00 00 00 00 00 00 08 00 27 11 22 33 00 00 00 00
0050 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
0060 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
0070 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
0080 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
0090 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00a0 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00b0 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00c0 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00d0 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00e0 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00f0 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
0100 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
0110 00 00 00 00 00 00 63 82 53 63 35 01 05 01 04 ffc..
0120 ff ff 00 03 04 ac 18 3f 01 06 08 ac 18 3f 0a 08?..
0130 08 08 08 0c 21 73 65 72 76 69 64 6f 72 2d 69 6dlser
0140 70 72 65 73 73 69 6f 2d 73 68 61 69 6c 61 6d 61 pressio-
0150 72 74 69 6e 65 7a 0f 14 73 68 61 69 6c 61 6d 61 rtinez.l
0160 72 74 69 6e 65 7a 2e 6c 6f 63 61 6c 2a 04 ac 18 rtinez.l
0170 3f 01 33 04 00 00 0e 10 36 04 ac 18 3f 0a 3a 04 ?-3;...
0180 00 00 07 08 3b 04 00 00 0c 4e 3d 07 01 08 00 27
0190 11 22 33 ff b0 04 00 00 0c 4e 3d 07 01 08 00 27 .."3..

Dynamic Host Configuration Protocol: Protocol Paquetes: 443 - Displayed: 2 (0.5%) Perfil: Default